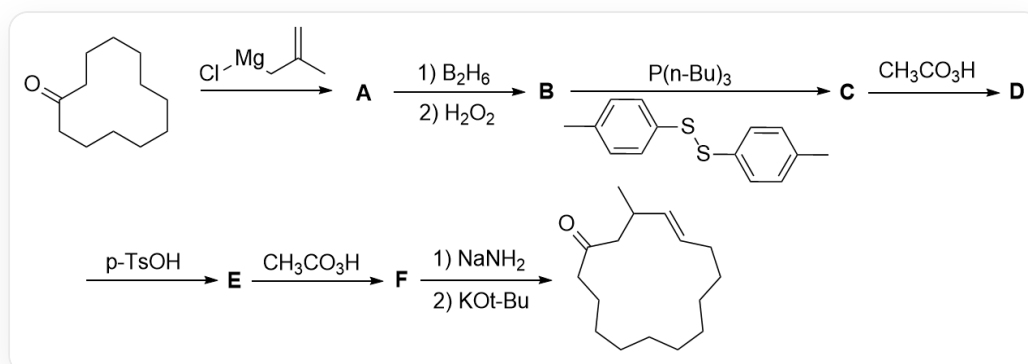


Question

The construction of macrocycles can be achieved through fragmentation reactions:



Reaction of C1CCCCC(=O)CCCC1 with C=C(C)C[Mg]Cl yields **A**, followed by reaction first with borane and then with hydrogen peroxide to yield **B**, reaction with tri-*n*-butylphosphine and CC1=CC=C(SSC2=CC=C(C)C=C2)C=C1 to yield **C**, reaction with peracetic acid to yield **D**, reaction with *p*-toluenesulfonic acid to yield **E**, followed by reaction with peracetic acid to yield **F**, and finally reaction first with sodium amide and then with potassium *tert*-butoxide to yield CC1/C=C/CCCCCCCCC(=O)C1

Select the correct item from the following options:

- A.** All other options are incorrect.
- B.** **B** contains two alcohol hydroxyl groups, both of which are tertiary alcohols.
- C.** **C** contains two oxygen atoms.
- D.** Both the generation of **D** and the generation of **F** reactions utilize peracetic acid. In both reactions, each molecule of reactant consumes one molecule of peracetic acid.
- E.** **E** contains 3 cycles

F. The generation of the final product 1 involves 3 cyclic intermediates.

Answer

Correct Answer: **F**

Detailed Explanation

First, the Grignard reagent C=C(C)C[Mg]Cl undergoes carbonyl addition to C1CCCCC(=O)CCCC1 to yield **A**: OC1(CC(C)=C)CCCCCCCCCCC1.

CHECKPOINT

1 PTS

The structure of **A** is OC1(CC(C)=C)CCCCCCCCCCC1

The generation of **B** from **A** is a typical hydroboration-oxidation condition, used to convert alkenes into alcohols. First, borane adds to the less sterically hindered carbon of the alkene, and is then converted to a hydroxyl group under the action of hydrogen peroxide, yielding **B**: OC1(CC(CO)C)CCCCCCCCCCC1, which has two alcohol hydroxyl groups, but one of them is a primary alcohol, so option B is incorrect.

CHECKPOINT

1 PTS

The structure of **B** is OC1(CC(CO)C)CCCCCCCCCCC1, which has two alcohol hydroxyl groups, but one of them is a primary alcohol, so option B is incorrect

When **B** is converted to **C**, first, tri-n-butylphosphine attacks the disulfide bond, producing CCCC[P+](CCCC)(CCCC)SC1=CC=C(C)C=C1, which then reacts with the least sterically hindered alcohol hydroxyl group, leaving thiophenol, to give OC1(CC(CO[P+](CCCC)(CCCC)CCCC)C)CCCCCCCCCCC1. Finally, thiophenol attacks,

leaving `O=P(CCCC)(CCCC)CCCC`, yielding **C**: `OC1(CC(CSC2=CC=C(C)C=C2)C)CCCCCCCCCCC1`, which has only one oxygen atom, so option C is incorrect.

CHECKPOINT

1 PTS

The structure of **C** is `OC1(CC(CSC2=CC=C(C)C=C2)C)CCCCCCCCCCC1`, which has only one oxygen atom, so option C is incorrect

Under the action of peracetic acid, the sulfur atom is oxidized, and the thioether is oxidized to a sulfone, giving **D**: `OC1(CC(CS(C2=CC=C(C)C=C2)(=O)=O)C)CCCCCCCCCCC1`

CHECKPOINT

1 PTS

The structure of **D** is `OC1(CC(CS(C2=CC=C(C)C=C2)(=O)=O)C)CCCCCCCCCCC1`

Under the action of a strong acid, the tertiary alcohol undergoes elimination, which can produce two alkenes **E1**: `CC(CS(C1=CC=C(C)C=C1)(=O)=O)C/C2=C/CCCCCCCCCCC2` and **E2**: `CC(CS(C1=CC=C(C)C=C1)(=O)=O)C=C2CCCCCCCCCCC2`. In the next step, the double bond is oxidized to an epoxide by peracetic acid, corresponding to **F1**: `CC(CS(C1=CC=C(C)C=C1)(=O)=O)CC2(O3)C3CCCCCCCCCCC2` and **F2**: `CC(CS(C1=CC=C(C)C=C1)(=O)=O)C(O2)C32CCCCCCCCCCC3`.

The generation of **D** consumes two molecules of peracetic acid, and the generation of **F** consumes one molecule of peracetic acid, so option D is incorrect.

CHECKPOINT

1 PTS

The generation of **D** consumes two molecules of peracetic acid, and the generation of **F** consumes one molecule of peracetic acid, so option D is incorrect

Under the action of a strong base, **F** loses a proton from the α -position of the sulfone, attacking the epoxide. At this time, **F2** cannot obtain a suitable fragmentation intermediate, only **F1** can produce `CC(C1S(C2=CC=C(C)C=C2)(=O)=O)CC3(O)C1CCCCCCCCC3`. **F1** has three rings, so option F is correct.

CHECKPOINT

1 PTS

The intermediate of the resulting product is `CC(C1S(C2=CC=C(C)C=C2)(=O)=O)CC3(O)C1CCCCCCCCC3`, which has three rings, so option F is correct

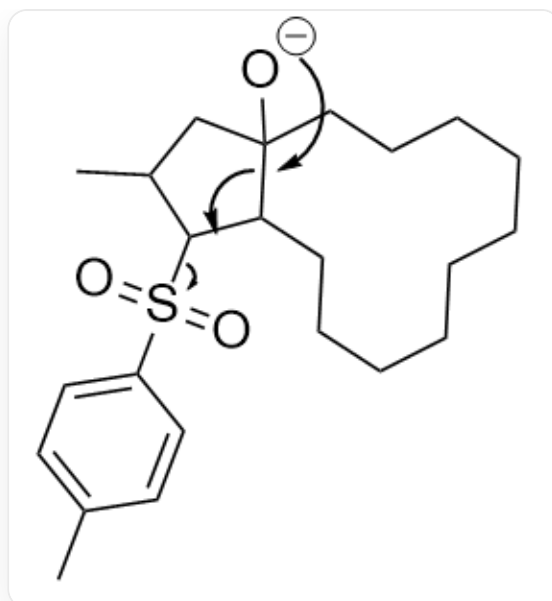
Therefore, the structure of **F** is `CC(CS(C1=CC=C(C)C=C1)(=O)=O)CC2(O3)C3CCCCCCCCC2`, and the structure of **E** is `CC(CS(C1=CC=C(C)C=C1)(=O)=O)C/C2=C/CCCCCCCCC2`, which has two rings, so option E is incorrect.

CHECKPOINT

1 PTS

The structure of **F** is `CC(CS(C1=CC=C(C)C=C1)(=O)=O)CC2(O3)C3CCCCCCCCC2`, and the structure of **E** is `CC(CS(C1=CC=C(C)C=C1)(=O)=O)C/C2=C/CCCCCCCCC2`, which has two rings, so option E is incorrect

Subsequently, the alcohol anion causes the newly formed five-membered ring to fragment, leaving `CC(C=C1)=CC=C1S([O-])=O`, to obtain the product.



Shows the fragmentation mechanism, where the negative charge of the alcohol anion is transferred to the connected carbon-carbon single bond, then a trans double bond is generated, leaving

